Lecture 4: Monopoly

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Lecture 3 Outline

- Monopoly revision
- Monopoly with two goods
  - Linked demands and linked costs
- Monopoly with two periods
  - Fixed durability and flexible durability
- Dominant Firm Model
  - Increasing, horizontal and combined fringe supply
- Further readings:
  - Belleflamme and Peitz (2015) Chapter 2.2.2, 2.2.3
  - Carlton and Perloff (1999) Chapter 15, 4
  - Bulow (1986) An economic theory of planned obsolescence. The Quarterly Journal of Economics
  - Church and Ware (2000) Chapter 4.2 (error in the figure)
  - **Notice:** Every textbook's model is slightly different (mainly in the assumption of the costs), and there might be some mistakes.



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What does a monopolist do

- A trade-off between q and p.
- Some usual assumptions: Downward sloping inverse demand function $p(q)$. Increasing cost function $c(q)$. Constant or upward sloping marginal cost $c^{'}(q)$.

$$max : \pi(q) = qp(q) - c(q)$$

$$\Longrightarrow FOD : qp^{'}(q) + p(q) - c^{'}(q) = 0$$

$$\Longrightarrow p(q) - c^{'}(q) = -qp^{'}(q)$$

$$\Longrightarrow \frac{p - c^{'}}{p} = \frac{-qp^{'}}{p} = -\frac{q}{p}\frac{\partial p}{\partial q} = -\frac{\frac{\partial p}{p}}{\frac{\partial q}{q}} = \frac{1}{\eta} = L \qquad (1)$$

- Monopoly pricing formula (Expression 1): mark up (Lerner index) equals to inverse elasticity of demand.



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What does a monopolist do

- Try to express the problem in the form of $q(p)$?
- When the absolute value of η decreases, it means the market demand is less affected by the price change, the monopolist is able to charge a higher price.

$$\frac{p - c^{'}}{p} = \frac{1}{\eta} \Longrightarrow p = \frac{c^{'}}{1 - \frac{1}{\eta}}$$

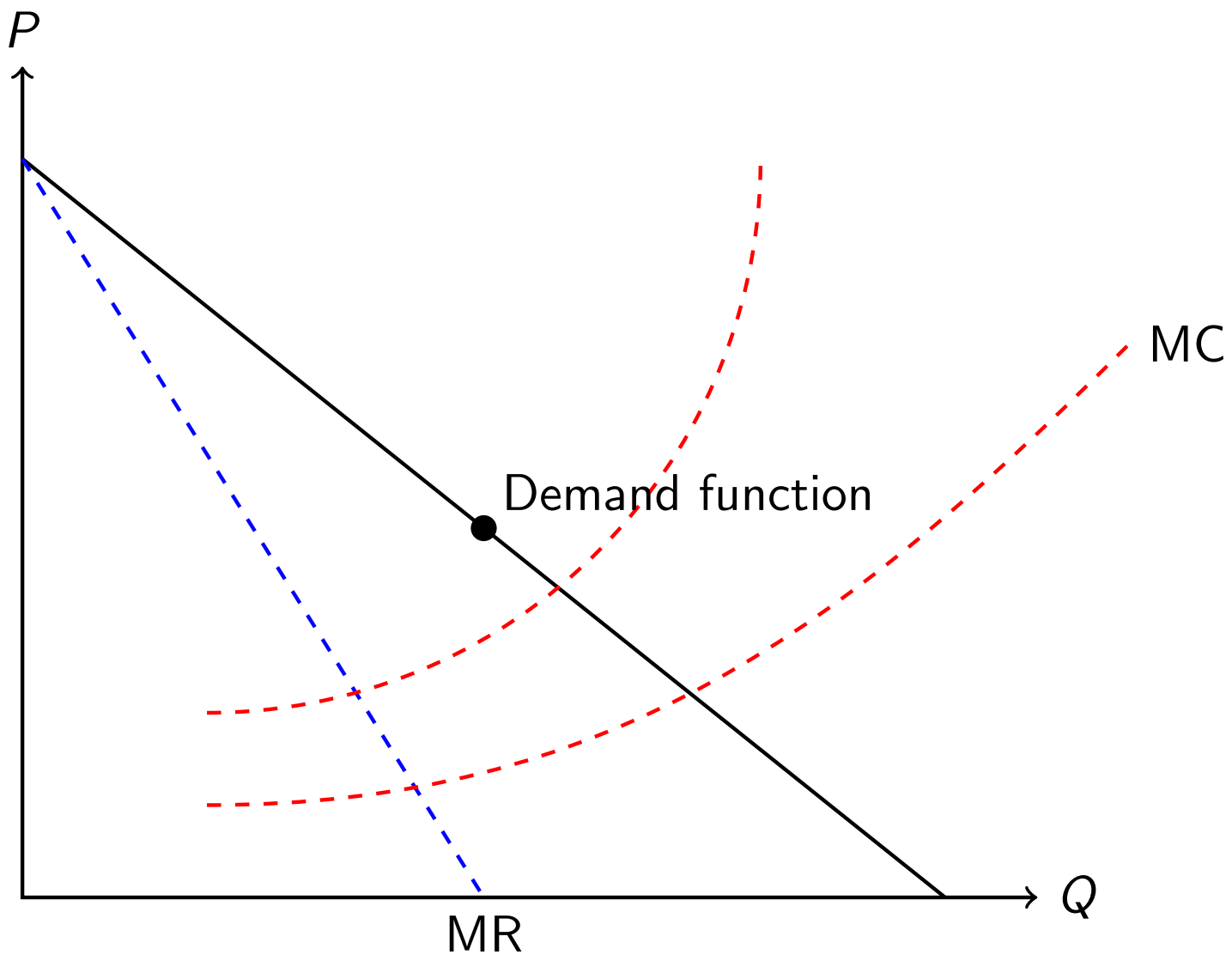
- How to interpret the extreme cases, say $\eta \to 1$ and $\eta \to \infty$?

$\eta \to \infty$	$\frac{p-c^{'}}{p} = 0$	$p \to c^{'}$, very small mark up
$\eta \to 2$	$\frac{p-c^{'}}{p} = \frac{1}{2}$	$p = 2c^{'}$, 100% mark up
$\eta \to 1$	$\frac{p-c^{'}}{p} = 1$	very high mark up

- Why we don't discuss $\eta < 1$?

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Monopoly, MR, MC, Elasticity



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Apple Inc.



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Monopoly pricing: two goods

- We now extend the earlier analysis to a multiproduct firm. We study the two-products case for simplicity.
- Demand function $q_1(p_1, p_2)$ and $q_2(p_1, p_2)$. Cost function $c(q_1, q_2)$.

$$max : \pi(p_1, p_2) = p_1 q_1(p_1, p_2) + p_2 q_2(p_1, p_2) - c(q_1(p_1, p_2), q_2(p_1, p_2)$$

$$\Longrightarrow FOD : q_i + p_i \frac{\partial q_i}{\partial p_i} + p_j \frac{\partial q_j}{\partial p_i} = \frac{\partial c}{\partial q_i} \frac{\partial q_i}{\partial p_i} + \frac{\partial c}{\partial q_j} \frac{\partial q_j}{\partial p_i} \quad (2)$$

note: $i \in \{1, 2\}, j \neq i$



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GPT Examples

Let's explore examples for each of the four scenarios you've described:

1. Linked Demand and Linked Costs

Smartphones and Proprietary Chargers: A company that manufactures a unique smartphone might also produce a proprietary charger that is required for the phone. The demand for the proprietary charger is directly linked to the demand for the smartphone as consumers need the charger to use the phone. The costs are linked because the same research, design, and possibly some components or materials are shared between the smartphone and the charger.

2. Linked Demand but Not Linked Cost

Razors and Razor Blades: A company might sell razors at a low price to encourage consumers to buy them, knowing that these consumers will need to continually purchase the company's specific razor blades. The demand for the blades is linked to the demand for the razor. However, the cost might not be linked because razors and razor blades could be produced using different materials, technologies, and processes.

3. Unlinked Demand but Linked Cost

Oil Refineries Producing Multiple Products: An oil refinery might produce both gasoline and asphalt. The demand for gasoline does not influence the demand for asphalt as they are used for different purposes (fuel vs. road construction). However, the costs are linked because the production of both comes from the same raw material—crude oil—and they share the costs of refining processes and infrastructure.

4. Unlinked Demand and Unlinked Cost

Book Publishing Company Producing Books and Bicycles: If a book publishing company also decided to manufacture bicycles, there would likely be no link between the demand for books and bicycles, as they serve entirely different consumer needs. Additionally, the costs are unlinked because the production processes, materials, and expertise required to create books are vastly different from those needed to manufacture bicycles.

These examples demonstrate the various relationships that can exist between demand and costs for products in different market situations. While linked demand and costs can offer competitive advantages and economies of scale, unlinked demand and costs may represent diversification strategies or unrelated business ventures..



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Linked and unlinked

- Unlinked demand (cost): it is possible to write the demand (cost) function of one of the good separately without involving the other good. Otherwise, it is linked.
- Unlinked Demand Functions: $q_1 = a - bp_1$ and $q_2 = d - fp_2$
- Linked Demand Functions: $q_1 = a - bp_1 + cp_2$ and $q_2 = d + ep_1 - fp_2$
- Unlinked Cost Functions: $c_1(q_1) = k_1 q_1^2$ and $c_2(q_2) = k_2 q_2^2$
- Linked Cost Function: $c(q_1, q_2) = k_1 q_1^2 + k_2 q_2^2 + k_3 q_1 q_2$



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Two goods: linked demands and unlinked costs

- When costs can be "separated" between the two activities: $c(q_1, q_2) = c_1(q_1) + c_2(q_2)$. Write $\frac{\partial c}{\partial q_i}$ as $c_i^{'}$ for simplicity, Expression 2 can be rewritten as

$$q_i + (p_i - c_i^{'})\frac{\partial q_i}{\partial p_i} + (p_j - c_j^{'})\frac{\partial q_j}{\partial p_i} = 0$$

- Dividing both sides by $p_i \frac{\partial q_i}{\partial p_i}$:

$$\frac{q_i}{p_i}\frac{\partial p_i}{\partial q_i} + \frac{p_i - c_i^{'}}{p_i} + \frac{p_j - c_j^{'}}{p_i}\frac{\partial q_j}{\partial p_i}\frac{\partial p_i}{\partial q_i} = 0$$



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Two goods: linked demands and unlinked costs

- Let $\eta_i = -\frac{p_i}{q_i}\frac{\partial q_i}{\partial p_i}$ denote the own elasticity of product i's demand, one obtains

$$L_i = \frac{p_i - c_i^{'}}{p_i} = \frac{1}{\eta_i} + \frac{p_j - c_j^{'}}{p_i}\frac{\frac{\partial q_j}{\partial p_i}}{-\frac{\partial q_i}{\partial p_i}} \tag{3}$$

- Compare Expression 3 with 1, the former expression has an additional term on the right-hand side. The term has the sign of $\frac{\partial q_j}{\partial p_i}$ (Notice that $\frac{\partial q_i}{\partial p_i}$ is negative and $p_j - c_j^{'}$ is positive.).



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Two goods: linked demands and unlinked costs

- When the two products are substitutes ($\frac{\partial q_j}{\partial p_i} > 0$), $L_i > \frac{1}{\eta_i}$. The multiproduct monopolist internalizes the competition effect between the two products: charging higher price for one product leads to a reduction of selling here, but an increase of selling there. This gives the monopolist an extra incentive to raise the price. Therefore, the multiproduct monopolist sets higher prices than separate divisions.
- When the two products are complements ($\frac{\partial q_j}{\partial p_i} < 0$), $L_i < \frac{1}{\eta_i}$. Reducing one price promotes two selling.

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Two goods: Half-time break

- So, if you are a monopolist with substitute or complementary products, what are your possible strategies?
- How to explain the free stuff consumers often receive: for example, free software, video game demos, etc?



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Two goods: unlinked demands and linked costs

- Suppose now that the demands are "separated" as $q_1(p_1)$ and $q_2(p_2)$, while the cost function has the general term $c(q_1, q_2)$. As the demands are independent, $\frac{\partial q_j}{\partial p_i} = 0$, Expression 2 becomes

$$L_i = \frac{p_i - c_i^{'}(q_i, q_j)}{p_i} = \frac{1}{\eta_i} \tag{4}$$

- Compare Expression 4 with Expression 1, there is only one minor difference: the Lerner index for product i also depends on q_j.



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Two goods: unlinked demands and linked costs

- Suppose there is Economies of Scope (cost advantage when produce a variety of products, compare it with economies of scale): $\frac{\partial^2 c(q_i,q_j)}{\partial q_i \partial q_j} < 0$, then producing $q_j \Longrightarrow c_i^{'}(q_i, q_j)$ decreases $\Longrightarrow$ the optimal p_i decreases. Hence, in the presence of economies of scope, the multiproduct monopolist has an incentive to set lower prices than separate firms. Similarly, with diseconomies of scope, the monopolist wishes to set higher prices and produces/sells less.
- Simple intuition: when producing one product more makes it cheaper to produce the other, why not?

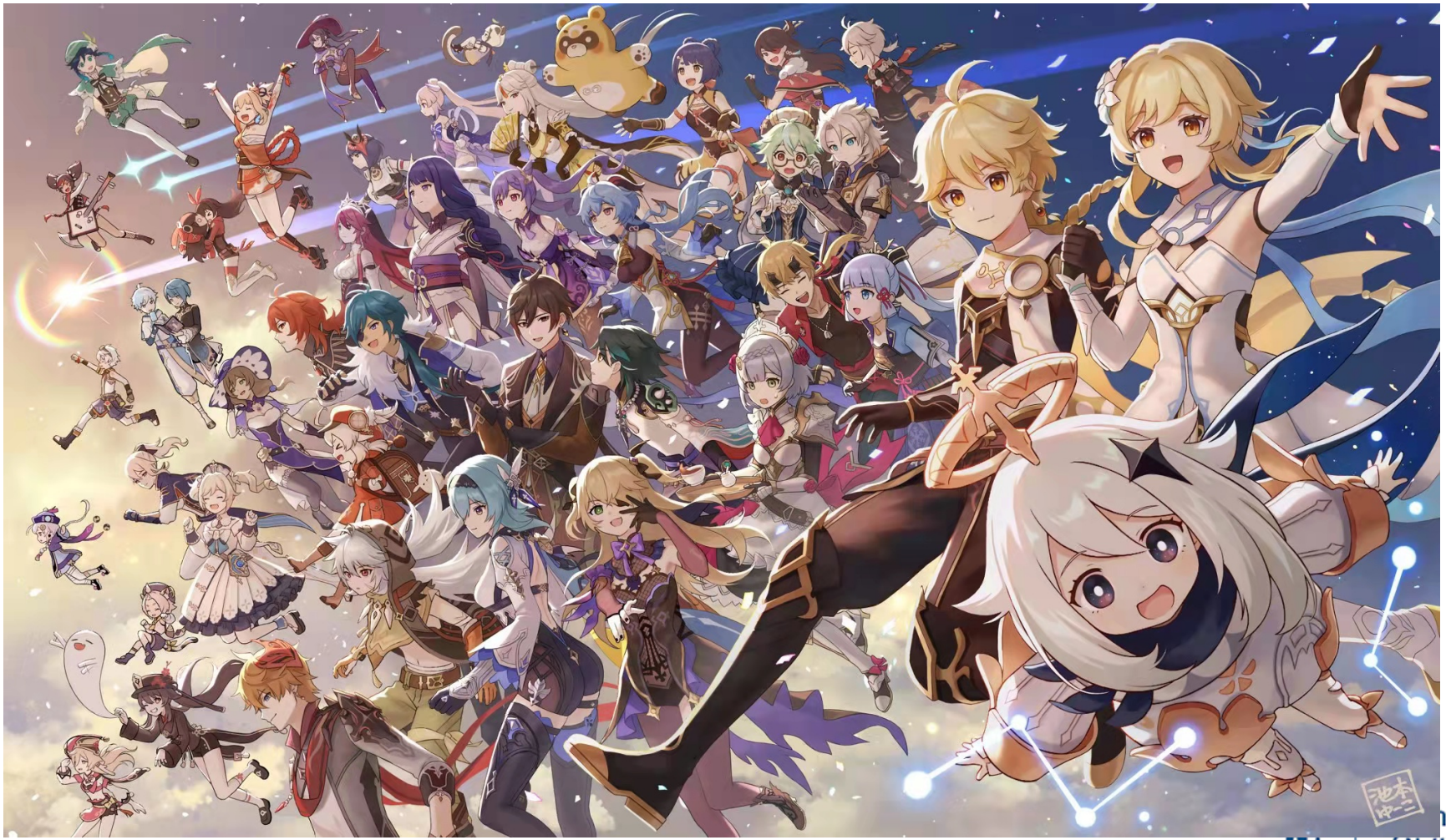


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Two goods: conclusion

- A multiproduct firm that has monopoly power over several products sets lower prices than separate firms (each controlling a single product) when the product are complements or when there are economies of scope among the products. It sets higher prices when the goods are substitutes or when there are diseconomies of scope.

Yuanshen (Genshin)



Monopoly pricing: two periods

- Now, instead of a firm producing two different goods at one period, we study a firm producing one good in multiple periods. Again, we study the two-periods case for simplicity.



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Monopoly pricing: two periods

- "duration": once a product is purchased, it could possibly be used for several (non-integer, for example, 1.5) periods. Alternatively: a number of products being purchased, some could be used for more than one period.
- Assume that of q_1 unit of product sold in the first period, $(1-\delta)q_1$ disappear at the end of the first period and δq_1 remain in period 2.

Monopoly pricing: two periods

- We start by considering the FIXED, PERFECT durability $\delta = 1$, so all of q_1 would still be in the market in the second period.
- The firm further produces an additional q_2 units in period 2.
- The **implicit rental price** (the willingness to pay to use the good for one period) in period 1 is $f_1(q_1)$ and the implicit rental price in period 2 is $f_2(q_1 + q_2)$. There is no cost for production. Notice that both implicit rental prices (f_1 and f_2) decreases with the market provision.
- Now, what is the selling prices for q_1 and q_2?



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Monopoly pricing: two periods

- Two ways to interpret the maximization problem:
- Renting: The monopolist tries to

$$max : \pi(q_1, q_2) = q_1 f_1(q_1) + (q_1 + q_2) f_2(q_1 + q_2)$$

- Selling: The monopolist tries to

$$max : \pi(q_1, q_2) = q_1(f_1(q_1) + f_2(q_1 + q_2)) + q_2 f_2(q_1 + q_2)$$

- Different? Why not?
- In the following pages we focus on the selling mode because renting is too simple. :)



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Implementing the selling plan

- Suppose that the monopolist announces the selling plan in advance and he implements the plan.
- The plan should satisfy the following first-order conditions:

$$\frac{\partial \pi}{\partial q_1} = f_1 + f_2 + q_1(f_1^{'} + f_2^{'}) + q_2 f_2^{'} = 0$$

$$\frac{\partial \pi}{\partial q_2} = f_2 + (q_1 + q_2) f_2^{'} = 0$$

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Not implementing the selling plan

- Now suppose that the monopolist does not announce the selling plan. (Even if he does, will he implement it?)
- What is his profit function in the second period? Will he try to maximize π?
- Actually, he would be now be thinking about

$$max : \pi^{\#}(q_2) = q_2 f_2(q_1 + q_2),$$

the solution of which implies that

$$\frac{\partial \pi^{\#}}{\partial q_2} = f_2 + q_2 f_2^{'} = 0$$

- Now that q_2 would be dependent on q_1, the original maximization function becomes

$$max : \pi(q_1) = q_1(f_1(q_1)+f_2(q_1+q_2(q_1)))+q_2(q_1)f_2(q_1+q_2(q_1))$$

- How would it affect the monopolist's decision?



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Numerical example

- Let's use a numerical example to better demonstrate. Suppose the implicit rental price function is unchanged over the two periods, $f = 20 - Q$. Q is the market provision.
- The monopolist tries to

$$max : \pi(q_1, q_2) = q_1(20 - q_1 + 20 - q_1 - q_2) + q_2(20 - q_1 - q_2)$$

- It is easy to derive that $q_1 = 10$ and $q_2 = 0$, $\pi(10, 0) = 200$. Correspondingly, $f_1 = 10$ and $f_2 = 10$. The profit-maximizing plan for the monopolist is to sell $q_1 = 10$ in the price of $p_1 = f_1 + f_2 = 20$ and does not sell q_2 at all.



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Numerical example

- However, if the monopolist carries out the plan and produces $q_1 = 10$, then in the second period, the monopolist would face the following maximization problem:

$$max : \pi^{\#}(q_2) = q_2(20 - q_1 - q_2).$$

- The FOD is $20 - q_1 - 2q_2 = 0$ and $q_2 = \frac{20-q_1}{2}$. Therefore, he WILL produce $q_2 = \frac{20-10}{2} = 5$ instead of $q_2 = 0$.
- It is easy to derive that $\pi(10, 5) = 175$. Less than 200! Does it seem counter-intuitive?

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Numerical example

- Notice that the selling price in period 1 p_1 is a combination of the implicit rental price $f_1(q_1)$ and $f_2(q_1 + q_2)$. It realizes only when consumers believe the producer to stick to the plan in the second period. Otherwise, consumers will not pay such a high price. In other words, if he produces $q_1 = 10$, the monopolist cannot convince consumers that he will no longer produce at the second period, and therefore would fail to sell at $p_1 = 20$.

Numerical example

- Anticipating this, what should be the q_1 for the monopolist, if he cannot convince the consumers that he will absolutely implement the selling plan?
- A "credible" q_2 has to be incentive compatible. Therefore, the profit function becomes: $q_2 = \frac{20-q_1}{2}$.

$$max : \pi(q_1, q_2(q_1)) = q_1(20-q_1)+(q_1+\frac{20-q_1}{2})(20-q_1-\frac{20-q_1}{2})$$

- It is easy to derive that $q_1 = 8$, $q_2 = 6$.
  $p_1 = f_1 + f_2 = 12 + 6 = 18$ and $p_2 = f_2 = 6$. $\pi(8, 6) = 180$.



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Numerical example

- In summarize, with a perfect reputation, the firm earns 200. Without the reputation and with a naive mind, the firm earns 175. Without the reputation but with a sophisticated mind, the firm earns 180. So how much do you think a reputation is worth here?
- Until now, we assume $\delta = 1$ so every product from the first period survives until the second period. If we assume otherwise $\delta < 1$, the calculation will be more difficult, but the intuition remains exactly the same.

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Two periods: flexible durability (optional)

- Now, what if the monopolist has control over the level of durability δ?
- Now the implicit rental price in period 1 is $f_1(q_1)$ and the implicit rental price in period 2 is $f_2(\delta q_1 + q_2)$. The costs in the two periods are $c_1(q_1, \delta)$ and $c_2(q_2)$. The monopolist tries to

$$max : \pi(q_1, q_2, \delta) = q_1 f_1(q_1) + (\delta q_1 + q_2) f_2(\delta q_1 + q_2)$$

$$- c_1(q_1, \delta) - c_2(q_2)$$



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Two periods: flexible durability (optional)

- Left unconstrained, the monopolist faces the following first-order conditions:

$$\frac{\partial \pi}{\partial q_1} = f_1 + q_1 f_1^{'} + \delta f_2 + (\delta q_1 + q_2)\delta f_2^{'} - \frac{\partial c_1}{\partial q_1} = 0$$

$$\frac{\partial \pi}{\partial q_2} = f_2 + (\delta q_1 + q_2) f_2^{'} - \frac{\partial c_2}{\partial q_2} = 0$$

$$\frac{\partial \pi}{\partial \delta} = q_1 f_2 + (\delta q_1 + q_2) q_1 f_2^{'} - \frac{\partial c_1}{\partial \delta} = 0$$



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Two periods: flexible durability (optional)

- Combining the last two conditions yields:

$$\frac{1}{q_1}\frac{\partial c_1}{\partial \delta} = \frac{\partial c_2}{\partial q_2} \tag{5}$$

Expression 5 states that the firm will choose an efficient durability: it equates the cost of making its original units a little more durable so that one more of the original units will still be useful in the second period, with the cost of replacing an extra unit in the second period.



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Two periods: flexible durability (optional)

- However, this analysis is incomplete and consequently incorrect. The problem of the durable goods monopolist is that the unconstrained solution to the maximization problem is generally not dynamically consistent. When the second period arrives, the monopolist will regard the first period as water under the bridge. It then faces the problem,

$$max : \pi^{'}(q_2) = q_2 f_2(\delta q_1 + q_2) - c_2(q_2),$$

the solution of which implies that

$$q_2 f_2^{'} + f_2 - c_2^{'} = 0$$

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Two periods: flexible durability (optional)

- Taking this into consideration, Expression 5 becomes (refer to the original paper for the very difficult derivation):

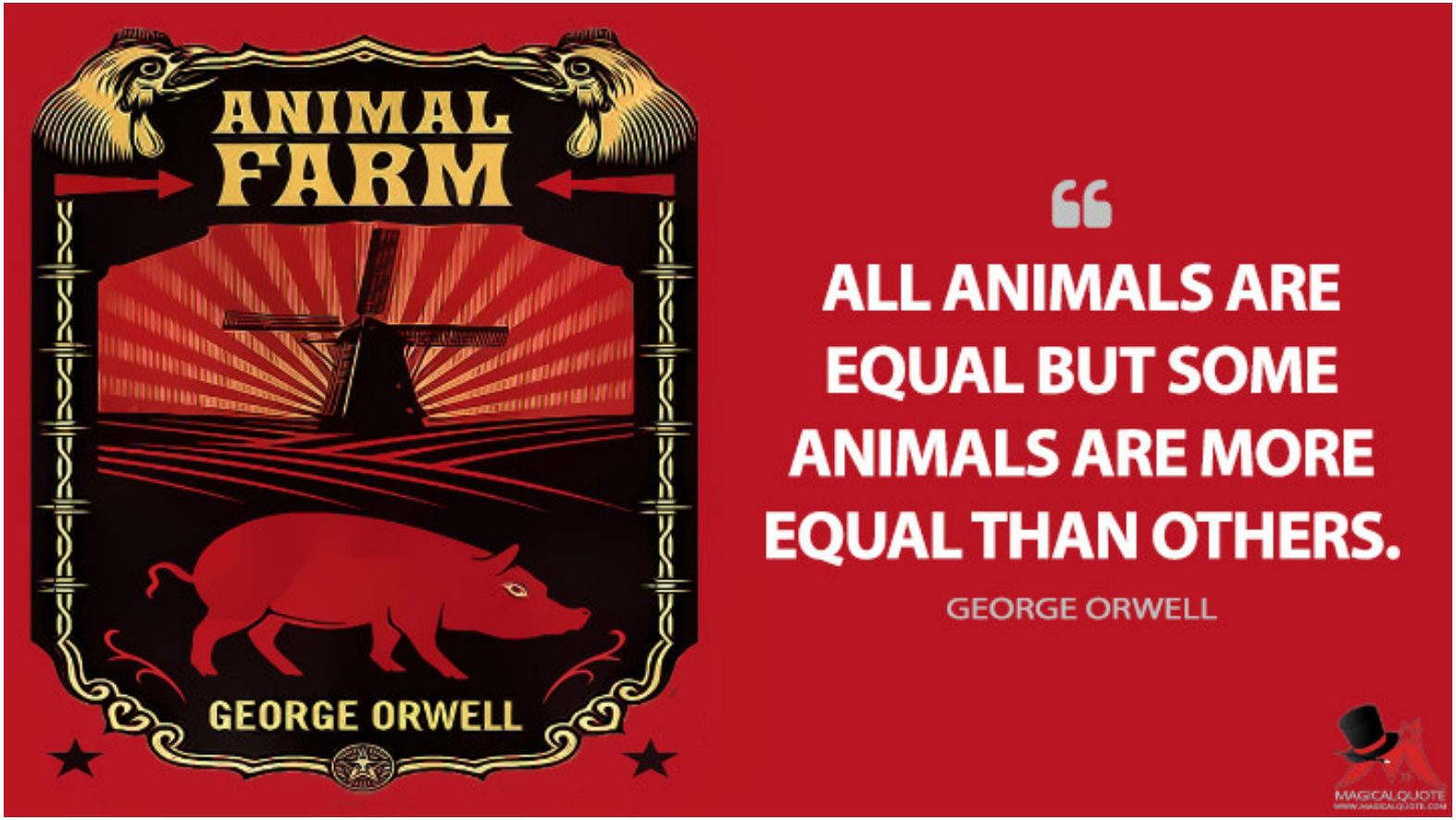
$$\frac{1}{q_1}\frac{\partial c_1}{\partial \delta} = c_2^{'} + \delta q_1 f_2^{'} \frac{d(\delta q_1 + q_2)}{d(\delta q_2)}$$

- If the marginal cost curve is steeper than the demand curve, an increase in δq_1 leads to a decrease in $(\delta q_1 + q_2)$, and the monopolist therefore chooses to produce too durable a product. Contrarily, the monopolist has an incentive to reduce durability below the efficient level, which is the "planned obsolescence".



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Why some firms are dominant



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Dominant Firm Model

What is so special about the dominant firm model:

- How a market might work when there is one very large firm with monopoly power, and many small firms acting as price-takers.
- It is a "deterministic" model, where most firms follow mechanical rules, in contrast to models involving games and strategies.
- To some extent, it is like a transition from Monopoly to Oligopoly.



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Model Assumption

- One large firm supplies most of the output, and can significantly affect the price through its choice of output level.
- Many smaller "fringe" firms also participate, but cannot influence the price, so are price-takers.
- Denote q_d as the dominant firm's output, q_f as the total fringe firms' output, and $q_m = q_d + q_f$ as the total market output.
- Market demand is

$$p = a - bq_m = a - b(q_d + q_f) \tag{6}$$

- We study the market outcome with different fringe supplies.



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Timeline

- Think of it like this: the dominant first chooses a q_d, which generates a temporary market price. Then the fringe firms act accordingly, produce q_f in total and drives the temporary market price to the final market price.
- When analyzing, it would be easier to do the backward induction: think about the final market price first, then, to achieve that final market price, what would the dominant firm's provision be?

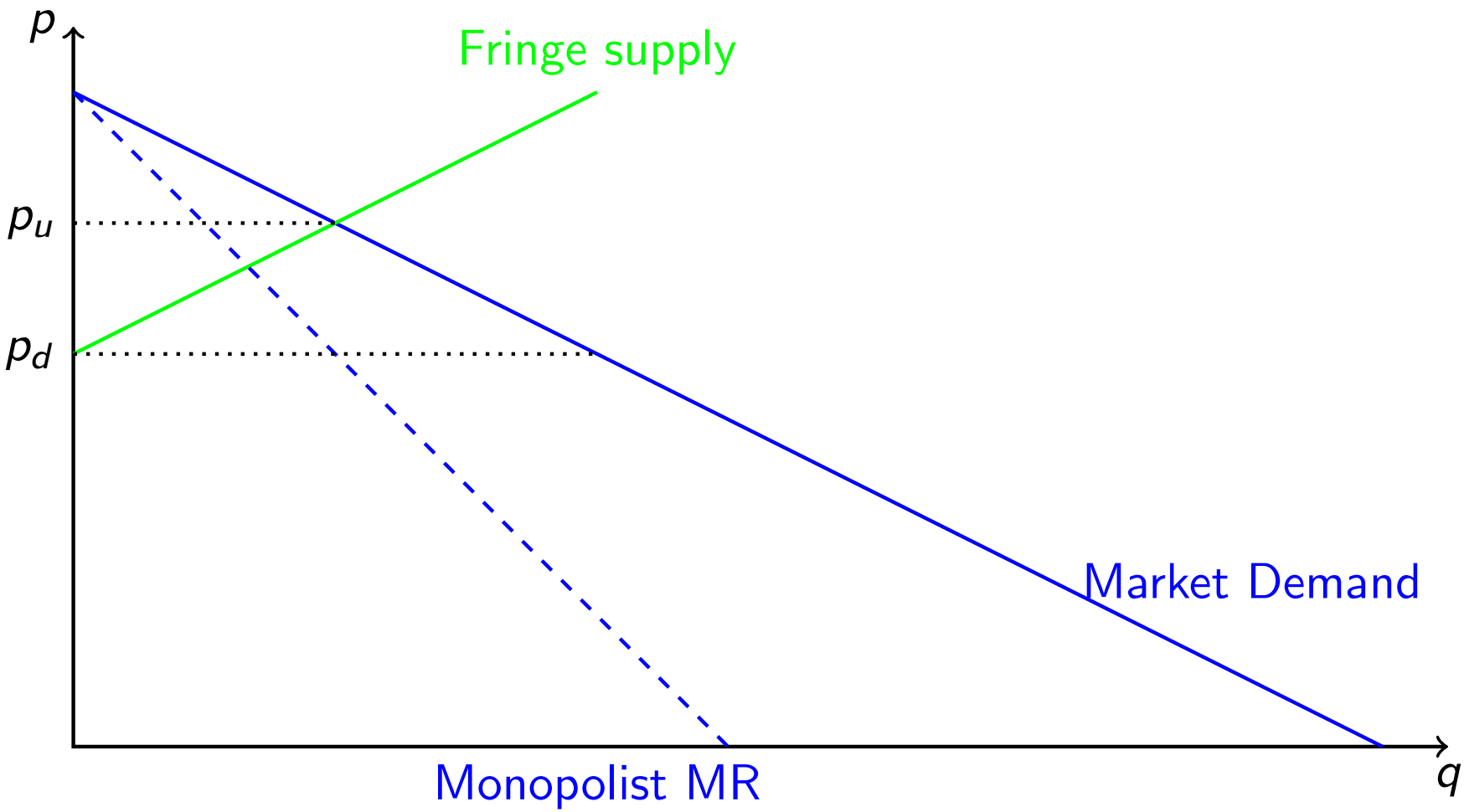
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Fringe supply I (increasing)

- Fringe supply is $p = d + eq_f \Longrightarrow q_f = \frac{p-d}{e}$.
- From the graph, let's think:
  - What happens when the final market price is above p_u?
  - What happens when the final market price is below p_d?
  - What happens when the final market price is between p_u and p_d?

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Fringe supply I (increasing)



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Fringe supply I (increasing)

When the final market price is between p_u and p_d:

- Combine with Expression 6. Substitute and eliminate q_f, we have

$$p = a - b(q_d + \frac{p-d}{e})$$

$$\implies p(1+\frac{b}{e}) = a - b(q_d - \frac{d}{e})$$

$$\implies p(e+b) = ea + bd - beq_d$$

$$\implies p(q_d) = \frac{ea+bd}{e+b} - \frac{be}{e+b}q_d,$$

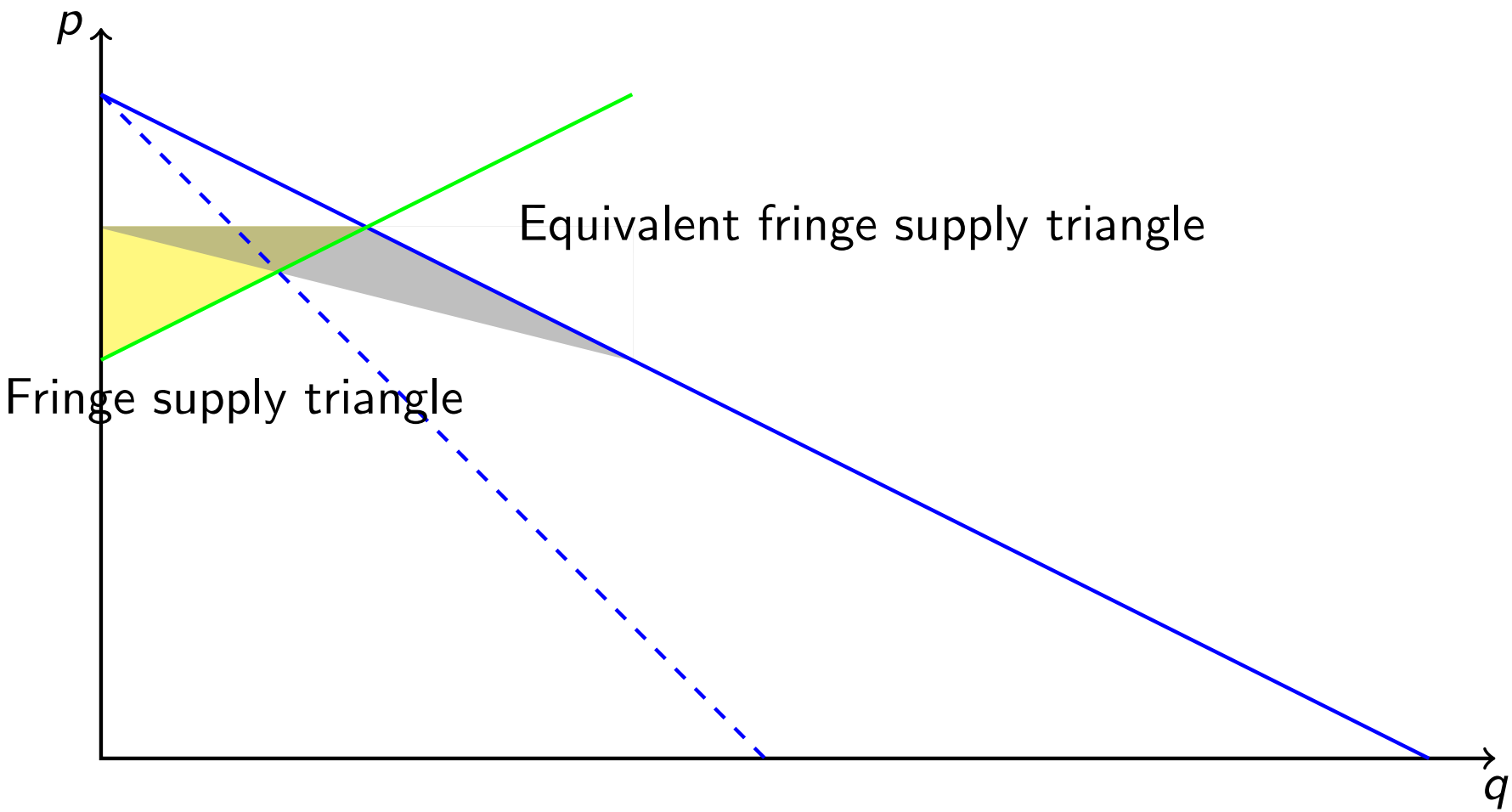
  which is the dominant firm's residual demand.

- Denote $\alpha = \frac{ea+bd}{e+b}$ and $\beta = \frac{be}{e+b}$, we have $p(q_d) = \alpha - \beta q_d$. Therefore, $MR(q_d) = \alpha - 2\beta q_d$.



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Fringe supply I (increasing)



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Fringe supply I (increasing)

Therefore, the residual demand is:

- $p(q_d) = \alpha - \beta q_d$ when $q_d < \frac{a-d}{b}$,
- $p(q_d) = a - bq_d$ when $q_d > \frac{a-d}{b}$.

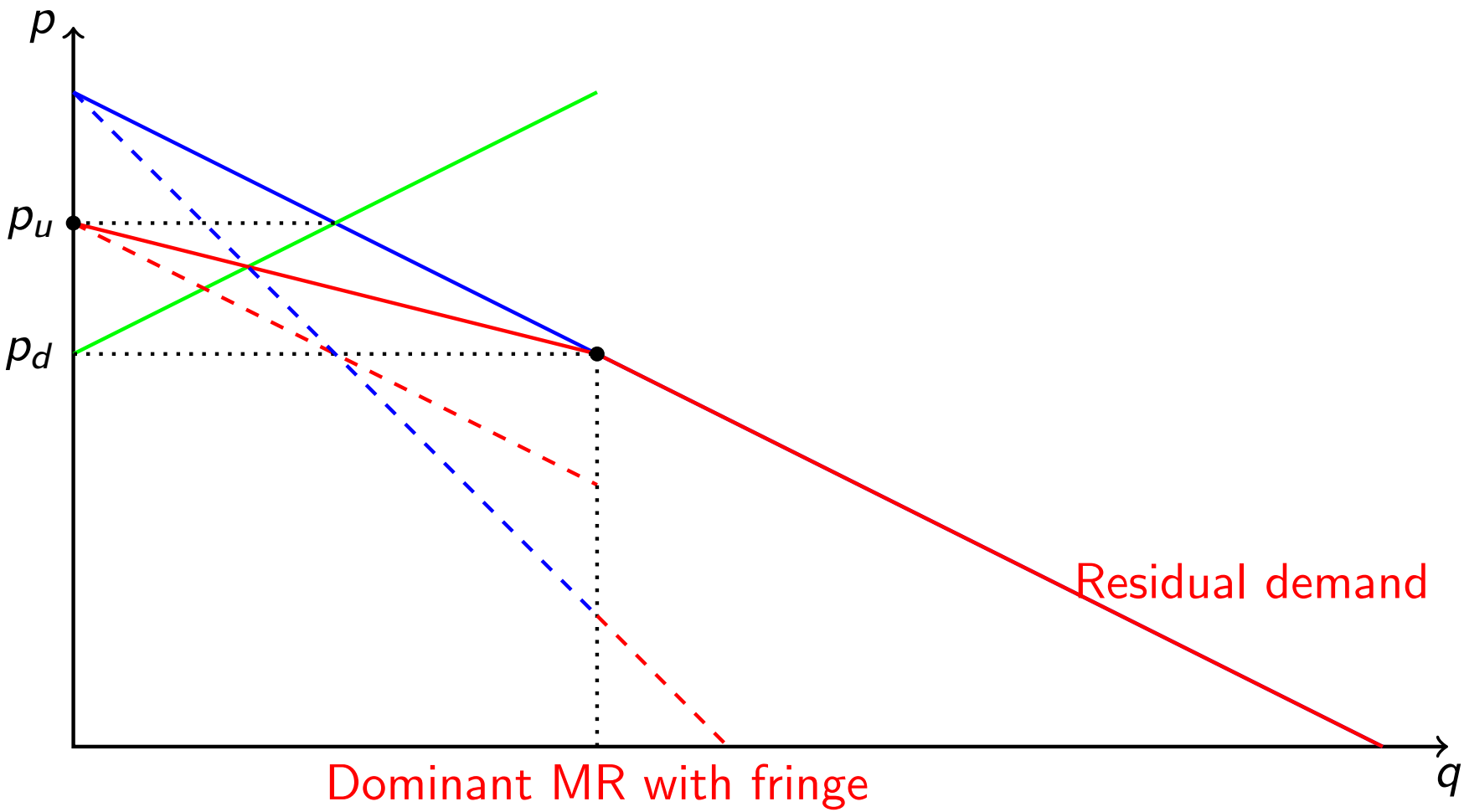
The dominant firm's marginal revenue is:

- $MR(q_d) = \alpha - 2\beta q_d$ when $q_d < \frac{a-d}{b}$,
- $MR(q_d) = a - 2bq_d$ when $q_d > \frac{a-d}{b}$.



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Fringe supply I (increasing)



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Fringe supply I (increasing)

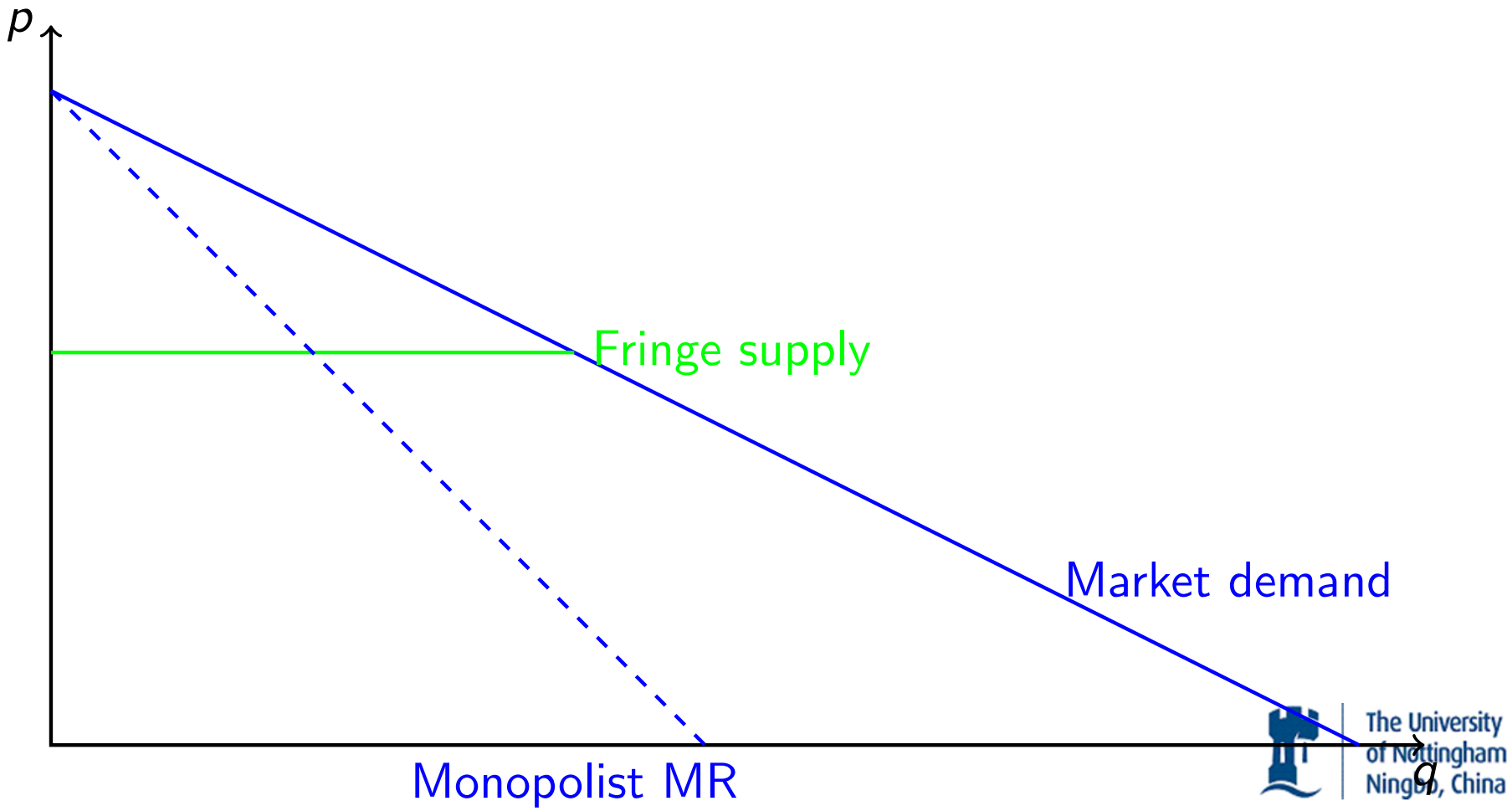
- What would the dominant firm's optimal decision-making be like?



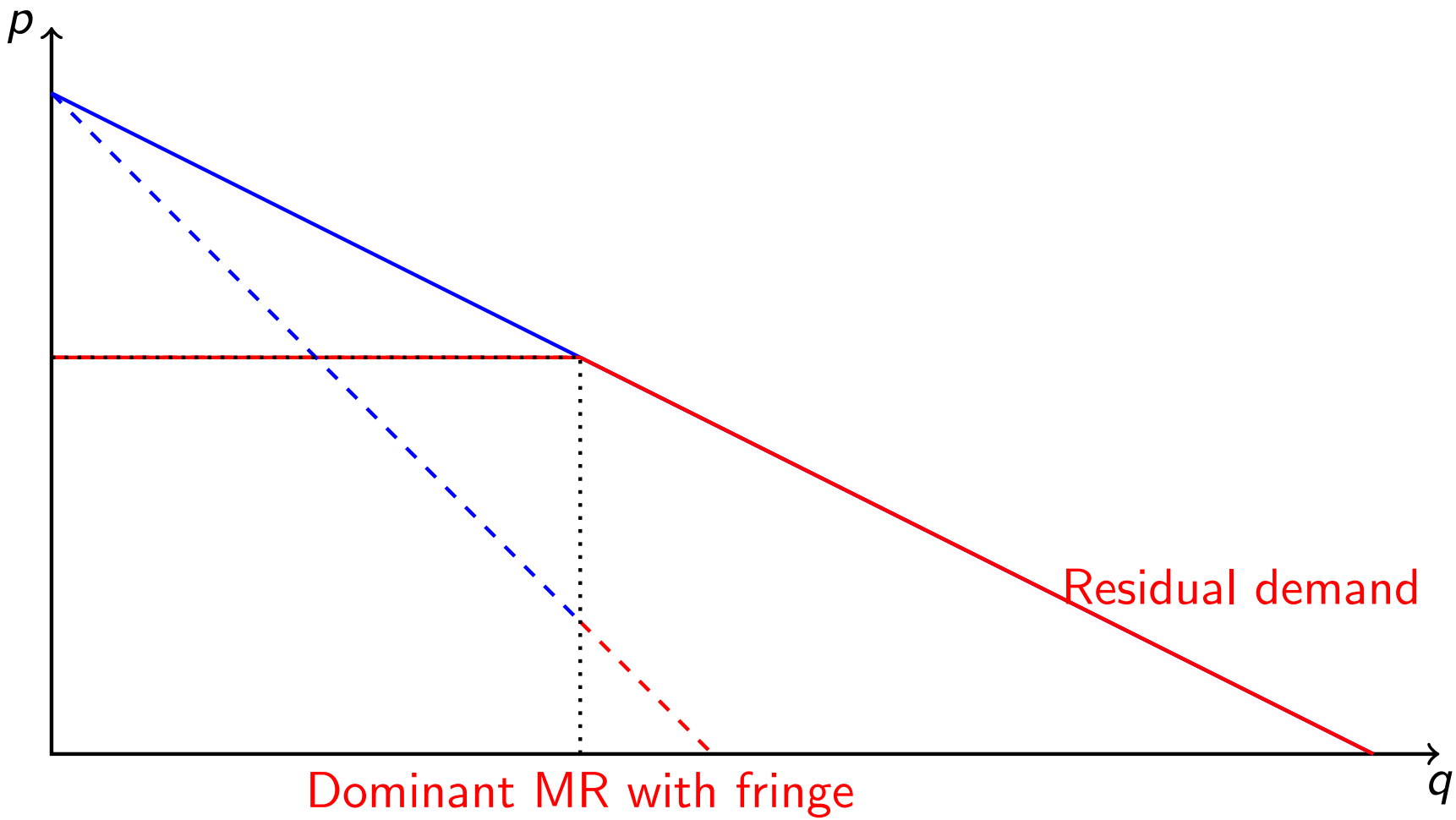
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Fringe supply II (horizontal)

- Now, think of a horizontal fringe supply: at or above a certain price, the fringe firms will provide as much as the market needs.



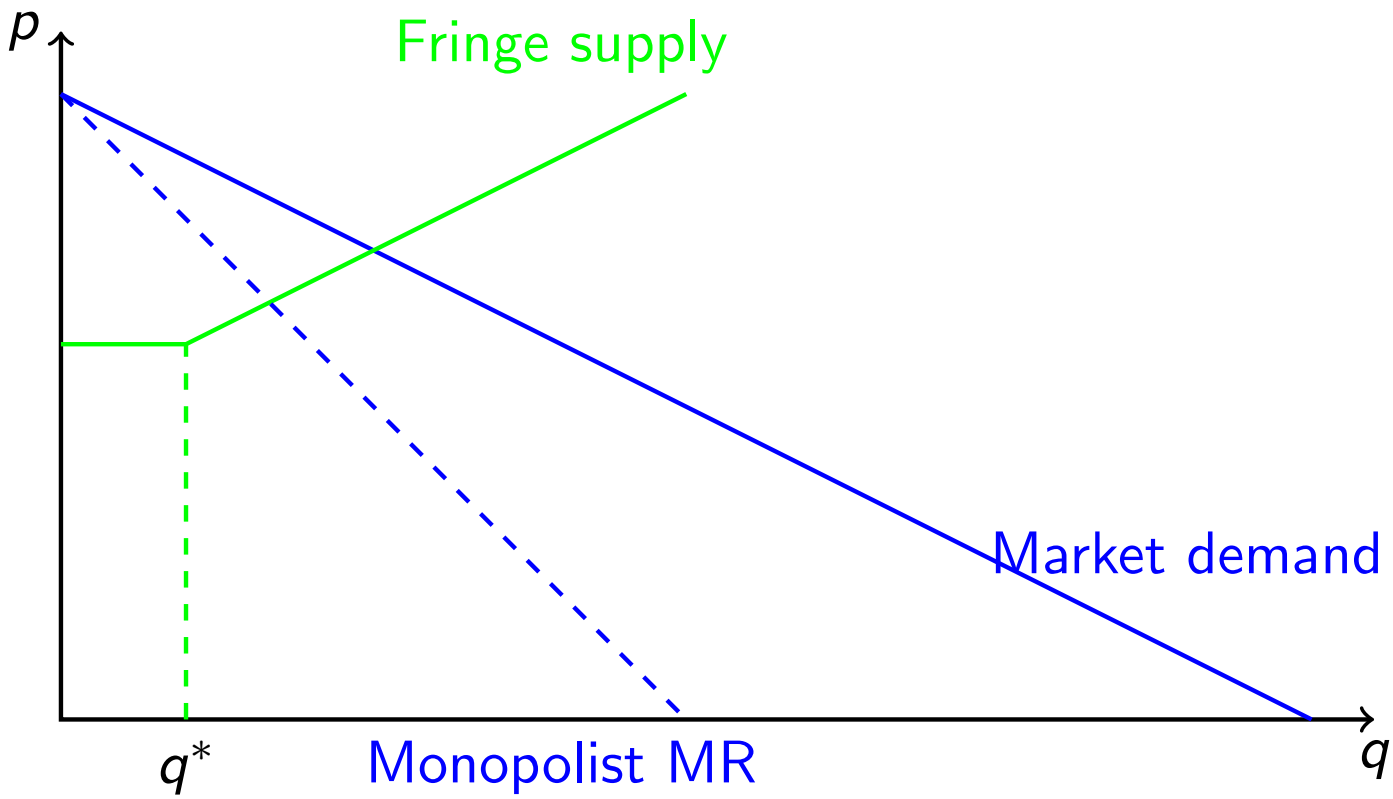
Fringe supply II (horizontal)



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Fringe supply III (combined)

- Now the fringe supply is like this:
  $p = \begin{cases} d & \text{when } q \leq q^* \\ d + e(q_f - q^*) & \text{when } q > q^* \end{cases}$. When $p = d$, fringe firms are willing to supply as much as q^*.



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Fringe supply III (combined)

- Similar to Model one, when the final market price is between p_u and p_d, substitute and eliminate q_f, we have

$$p = a - b(q_d + \frac{p-d}{e} + q^*)$$

$$\Longrightarrow p(1 + \frac{b}{e}) = a - b(q_d - \frac{d}{e} + q^*)$$

$$\Longrightarrow p(e+b) = ea - b(eq_d - d + eq^*)$$

$$\Longrightarrow p(q_d) = \frac{ea+bd}{e+b} - \frac{be}{e+b}q_d - \frac{be}{e+b}q^*,$$

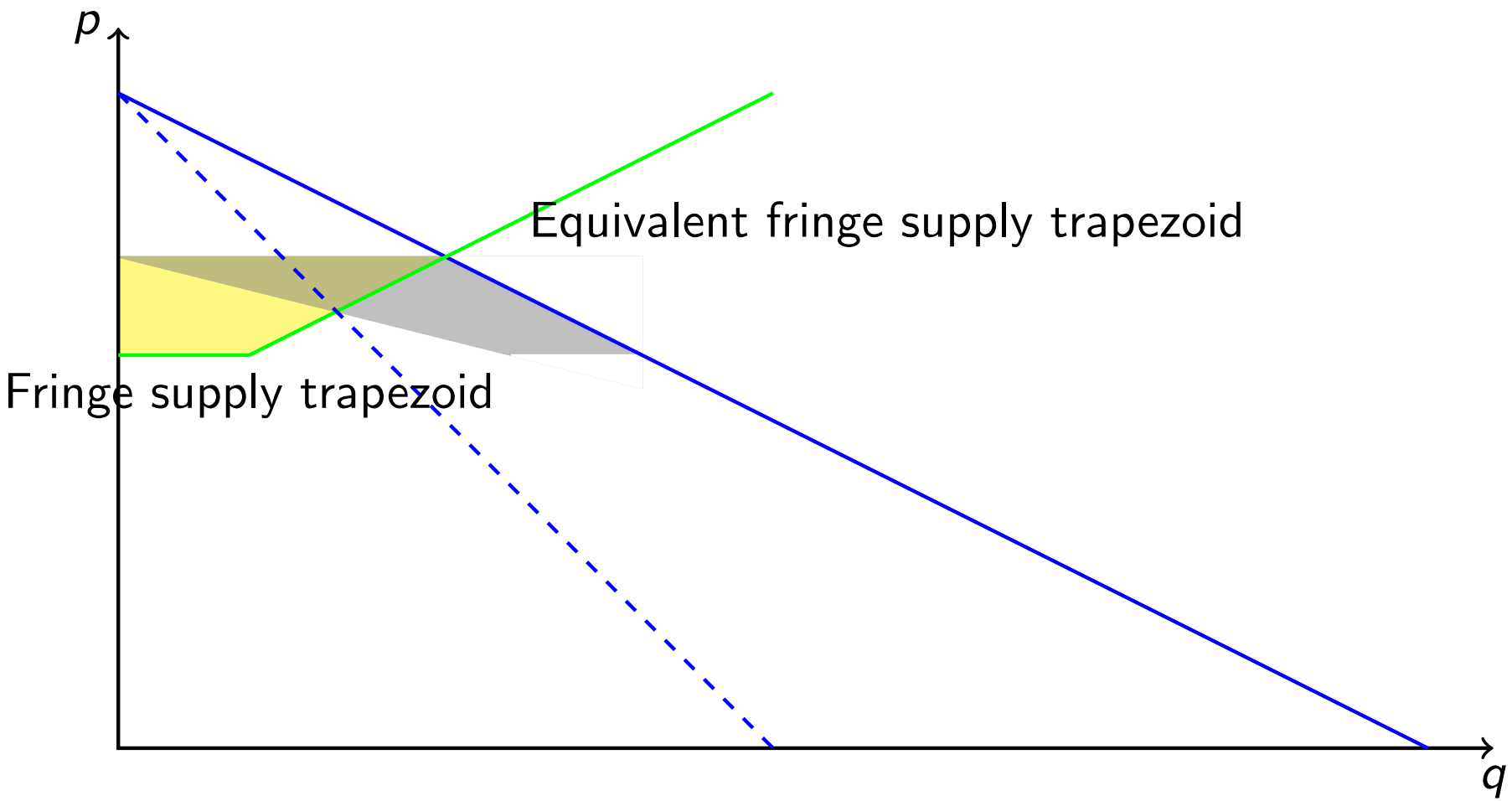
  which is the dominant firm's residual demand $p(q_d) = \alpha - \beta q_d - \beta q^*$. $MR(q_d) = \alpha - 2\beta q_d - \beta q^*$.

- The tricky question is: what about when $p = p_d$?



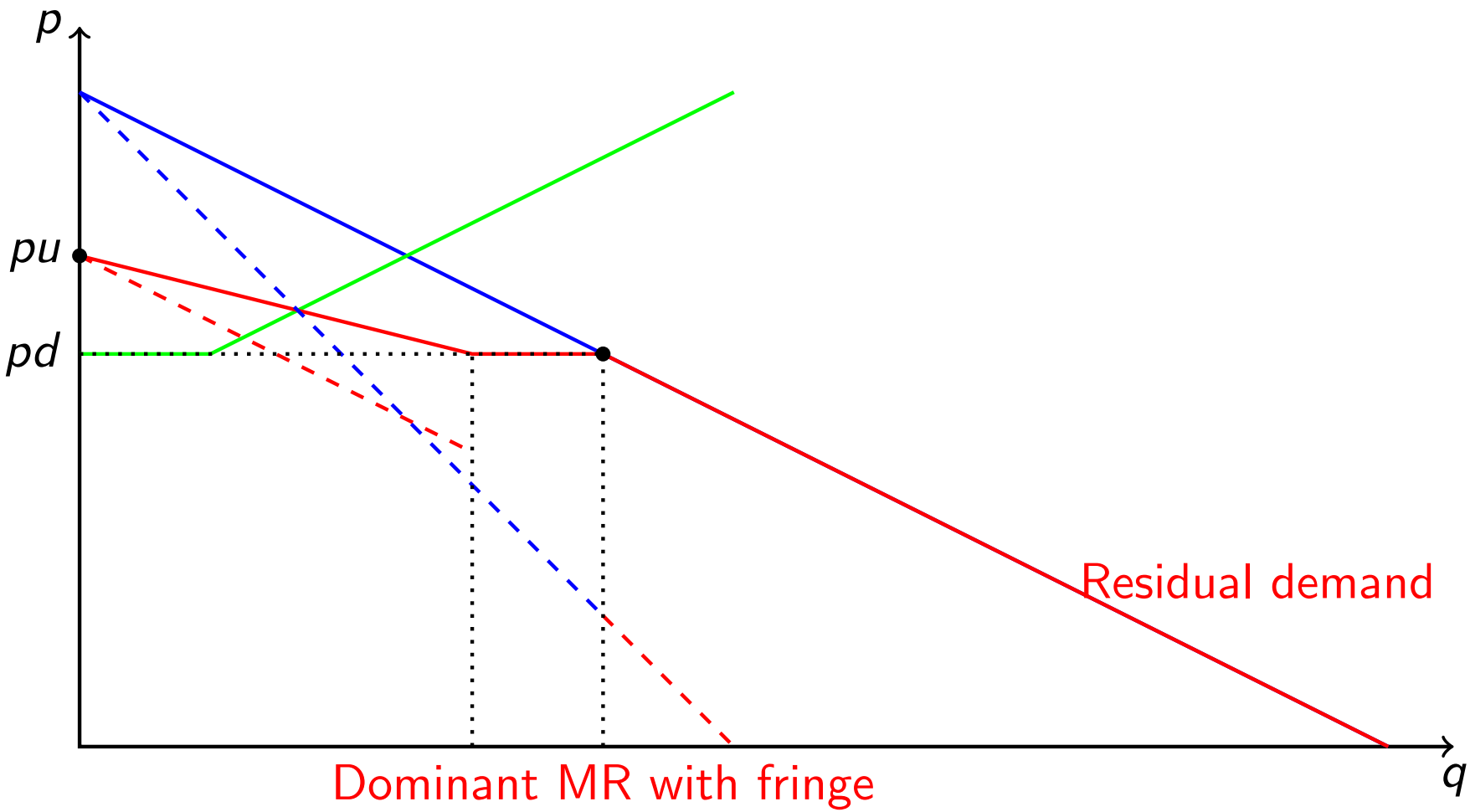
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Fringe supply III (combined)



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Therefore, the residual demand is:

- $p(q_d) = \alpha - \beta q^* - \beta q_d$ when $q_d < \frac{a-d}{b} - q^*$,
- $p(q_d) = d$ when $\frac{a-d}{b} - q^* < q_d < \frac{a-d}{b}$,
- $p(q_d) = a - bq_d$ when $q_d > \frac{a-d}{b}$.

The dominant firm's marginal revenue is:

- $MR(q_d) = \alpha - \beta q^* - 2\beta q_d$ when $q_d < \frac{a-d}{b} - q^*$,
- $MR(q_d) = d$ when $\frac{a-d}{b} - q^* < q_d < \frac{a-d}{b}$,
- $MR(q_d) = a - 2bq_d$ when $q_d > \frac{a-d}{b}$.



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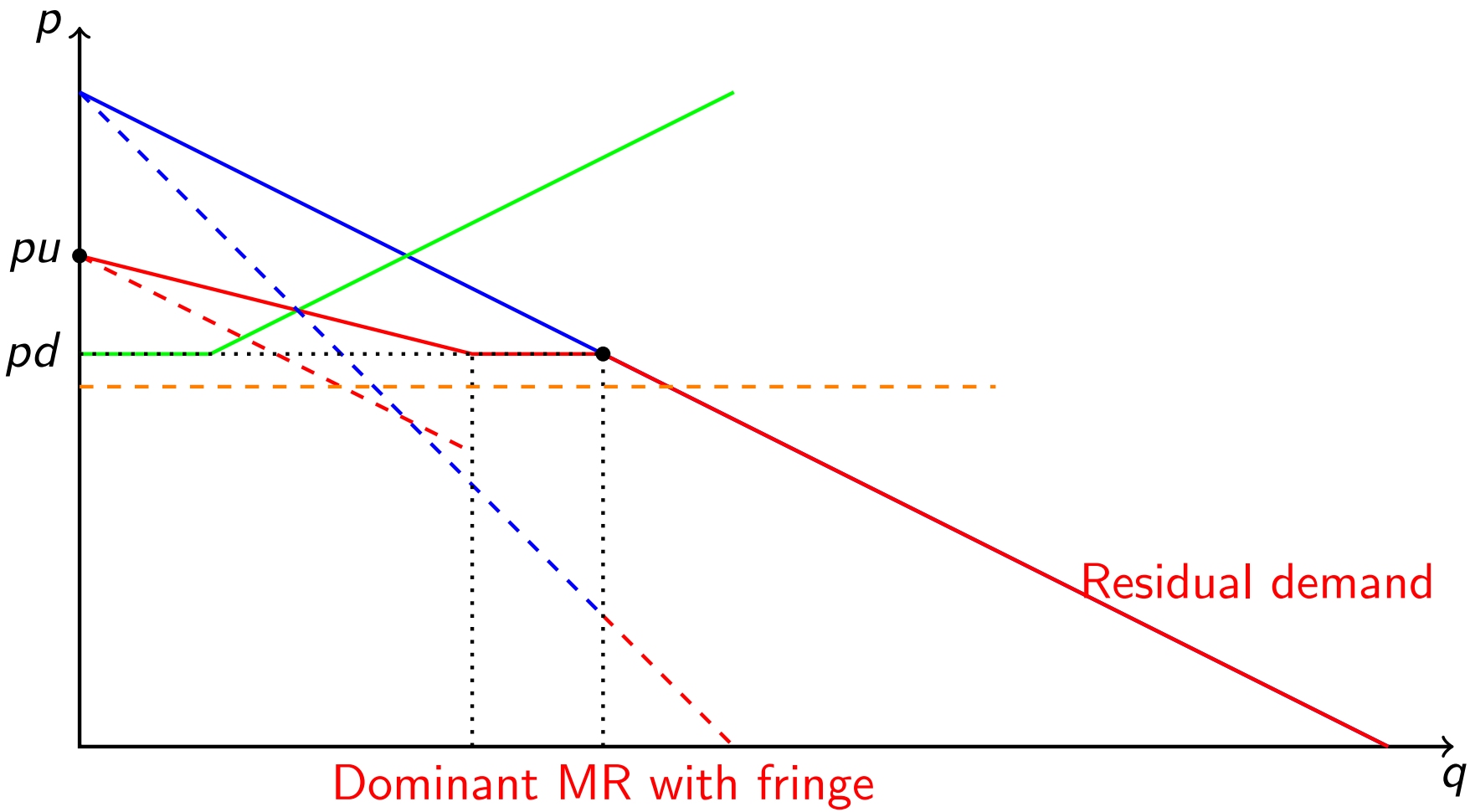
Fringe supply III (combined)

- What would the dominant firm's optimal decision-making be like?
- Intuition of the non-monotonicity?



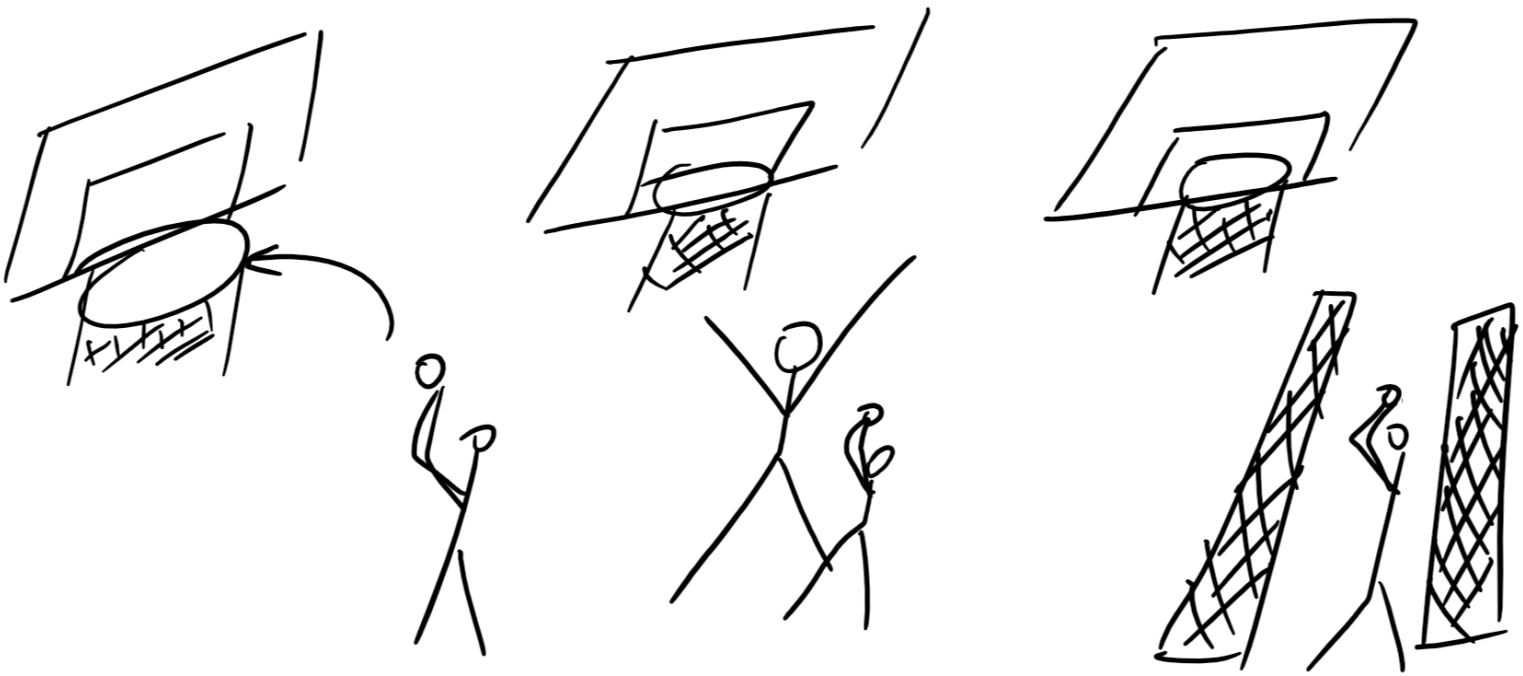
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Thank you



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